Aditya Amol Bhongade

On request → | aabhongade@ycce.edu | adityabhongade.github.io
Profile | adityabhongade | Google Scholar
Nagpur, Maharashtra, India

Experience

IIT (ISM) Dhanbad *Feb 2026 - Jul 2026*
Research Intern [Certificate]

- Under the supervision of **Dr. Partha Pratim Roy** (Dept. of CSE).
- **Field of Work:** Vision Language Models for Complex Document Understanding.
- **Skills:** Deep Learning, Large Language Models (LLMs), Vision Language Models,
- **Technologies:** PyTorch, Hugging Face, HF Transformers and Diffusers, TRL, vLLM, llama.cpp.

Education

Yeshwantrao Chavan College of Engineering *Nov 2022 - Jun 2026*
Bachelor of Technology (B. Tech) in Electronics & Telecommunication Engineering Nagpur, India

- CGPA: 8.8/10
- **Bachelor's Project:** "Deep Learning-based Liver Fibrosis Grading System with Conversational Support through Medical LLM" | [Code] [Demo] [Video]

Fellowship

IndiaAI Fellowship *May 2026*
Government of India

- Awarded with the prestigious IndiaAI Fellowship worth **INR 100K**.
- **Project Title:** "Energy based Face Matching Model for ID Card Verification"
- Media Coverage: Lokmat Times (English) | Lokmat (Marathi)

Publications

[J.1] Yogita, Dubey, Aditya Bhongade, et al. (2025). **LivSCP: Improving Liver Fibrosis Classification Through Supervised Contrastive Pretraining**. *Diagnostics*, Vol. 15, No. 24, Article 3226. DOI: 10.3390/diagnostics15243226.

[J.2] Aditya Bhongade, et al. (2025). **Robust and Explainable Ensemble Based Framework for Liver Disease Classification Using Data Balancing and Upsampling**. *SSRG International Journal of Electronics and Communication Engineering*, Vol. 12, Issue 2, pp. 1–11. DOI: 10.14445/23488549/IJECE-V12I2P101.

[J.3] Yogita Dubey, Aditya Bhongade, et al. (2024). **Efficient Explainable Models for Alzheimer's Disease Classification with Feature Selection and Data Balancing Approach Using Ensemble Learning**. *Diagnostics*, Vol. 14, No. 24, Article 2770. DOI: 10.3390/diagnostics14242770.

[C.1] Aditya Bhongade, et al. (2025). **Transformer and CNN-Based Lightweight Ensemble for Retinal Disease Classification Using OCT Imaging**. In *TENCON 2025 - 2025 IEEE Region 10 Conference (TENCON)*, pp. 1068–1072. DOI: 10.1109/TENCON66050.2025.11375147.

[C.2] Aditya Bhongade, et al. (2025). **LightSeg: A Lightweight Transformer Based Network for Lung Segmentation in Chest X-Rays**. In *2025 IEEE 6th India Council International Subsections Conference (INDISCON)*, pp. 1–6. DOI: 10.1109/INDISCON66021.2025.11253558.

[C.3] Aditya Bhongade, et al. (2025). **Explainable Vision Ensemble for Alzheimer's Disease Classification Using Brain MRI**. In *2025 IEEE 6th India Council International Subsections Conference (INDISCON)*, pp. 1–6. DOI: 10.1109/INDISCON66021.2025.11251953.

[C.4] Aditya Bhongade, et al. (2025). **An Efficient and Explainable Model for Heart Disease Classification Using Boosting Algorithms**. In *2025 6th International Conference for Emerging Technology (INCET)*, pp. 1–6. DOI: 10.1109/INCET64471.2025.11140243.

[C.5] Aditya Bhongade, et al. (2024). **Retinal Blood Vessel Segmentation Using Twin U-Net Based Deep Learning Architecture**. In *2024 IEEE 21st India Council International Conference (INDICON)*, pp. 01–06. DOI: 10.1109/INDICON63790.2024.10958322.

[C.6] Aditya Bhongade, et al. (2024). **Explainable AI Model for Medical Insurance Premium Prediction Using Ensemble Learning with SHAP Analysis**. In *2024 IEEE Pune Section International Conference (PuneCon)*, pp. 1–6. DOI: 10.1109/PuneCon63413.2024.10895118.

Honors and Awards

Ranker *Nov 2025 – Jan 2026*
Yeshwantrao Chavan College of Engineering

- 7th Semester: **1st rank** among 214 students (**SGPA 9.5/10**). Jan 2026
- 6th Semester: **3rd rank** among 214 students (**SGPA 9.41/10**). Nov 2025

Communication Skills Top Performer *Apr 2025*
Yeshwantrao Chavan College of Engineering

- Awarded a trophy and certificate for being a top performer in all aspects of Listening, Speaking, Reading, and Writing skills.

Funded Projects

SAR Image Colorization using Generative Adversarial Networks *Dec 2025 - Present*

- Received funding of INR 45,000.
- Designed novel generator and discriminator neural networks for SAR image colorization.

Profiles

LeetCode

- Profile
- Top 13% | Awarded with LeetCode T-Shirt | 30 Badges Earned | 990+ Problems Solved

Skills

Languages: C++, Python, SQL
Deep Learning & AI: PyTorch, Hugging Face, HF Transformers and Diffusers, CUDA, timm
DevOps & Version Control: Git, GitHub
Specialized Area: Deep Learning, Image Processing, Medical AI, Large Language Models, Vision Language Models
Mathematical & Statistical Tools: MATLAB
Research Skills: Literature Review, Academic Writing, LaTeX

Services

Publicity Master *Sep 2024 - Nov 2025*
Codeware Coding Club

- Responsible for leading the Publicity department of the club. Made decisions on the club's promotion and outreach policies.

Reviewer

- ***IEEE International Conference on Acoustics, Speech, and Signal Processing - ICASSP 2027*** *Sept 2026 - Present*
- ***17th Indian Conference on Computer Vision, Graphics and Image Processing - ICVGIP 2026*** *Sept 2026 - Present*
- ***IEEE 6th India Council International Conference (INDISCON) 2025*** *May 2025 - May 2025*

Invited/Tutorial Speaker

International Conference on Trustworthy AI and Intelligent IoT Systems (TAIIS 2026) *Dec 2026*
Asia University, Taiwan

- Deploying Trustworthy Medical VLMs: Reasoning, Retrieval, and Grounding

ISTE-approved Faculty Development Program (FDP) on Python & Data Science *Dec 2026*
Yeshwantrao Chavan College of Engineering, India

- Attention Mechanism & Applications in Healthcare

Summer/Winter Schools

Indian Institute of Science (IISc): KIAC Winter School on AI for FinTech *Jan 2026*

- Attended a 7-days winter school on AI and FinTech at IISc Bangalore, India.

Indian Institute of Science (IISc): EE Summer School *Jul 2025*

- Attended a 3-days winter school on AI and Signal Processing at IISc Bangalore, India.

Certifications

Cognitive Class by IBM: Machine Learning with Python *Apr 2025*

Coursera and University of Michigan: Python Programming for Everybody *Feb 2025*

Cambridge University Press: Cambridge University English Empower Level C1 *Feb 2025*